Journal of Science and Medicine in Sport 28 (2025) 587–593

Contents lists available at ScienceDirect

Journal of Science and Medicine in Sport

journal homepage: www.elsevier.com/locate/jsams

Original research

The role of generic cognitive skills: An empirical investigation into the association between generic and sport-specific cognitive skills and playing level in youth football

Martin L. Reinhard [a,b,*], David L. Mann [c], Oliver Höner [a]

[a] Eberhard Karls University, Institute of Sports Science, Department Sport Psychology and Research Methods, Germany
[b] VfB Stuttgart AG, Germany
[c] Vrije Universiteit Amsterdam, Department of Human Movement Sciences, Amsterdam Movement Sciences and Institute Brain and Behavior Amsterdam (iBBA), the Netherlands

ARTICLE INFO

Article history:
Received 12 August 2024
Received in revised form 23 December 2024
Accepted 24 January 2025
Available online 30 January 2025

Keywords:
Talent identification
Youth sport
Executive function
Decision making
Football intelligence
Regression analysis

ABSTRACT

Objectives: Tests of cognitive skills are increasingly used to identify talent in sports such as football, but there is so far little evidence to suggest that those tests are associated with sport-specific cognitive skills and performance. The aim of this study was to establish the degree to which tests of generic cognitive skills are associated with football-specific skill in youth football. Specifically, we (i) established the degree to which tests of generic cognitive skills are associated with sport-specific cognitive skills, and (ii) compared the associations between generic and sport-specific cognitive skills and the playing level of youth footballers.
Design: Cross-sectional study.
Methods: Youth football players (U11–U15; $N = 110$) competing at an elite and sub-elite level conducted two diagnostic tests of generic and one of football-specific cognitive skills. Generic tests were (1) the *determination test* measuring reactive stress tolerance, and (2) *NeurOlympics* assessing executive functions and anticipation to produce a *Football Intelligence* score (FI score). The football-specific test was a 360°-video decision-making test. Multiple regression analyses were used to establish the association between the tests, and logistic regression for their association to playing level (elite vs. sub-elite).
Results: The generic *Football Intelligence* score showed a small but significant association with football-specific decision making when controlling for age ($p = 0.006$). However, football-specific decision making (Nagelkerke's $R^2 = 0.34$, $p < 0.001$) had a higher diagnostic validity when predicting the playing level than did the FI score (Nagelkerke's $R^2 = 0.13$, $p < 0.001$).
Conclusions: We found a partial relationship between generic and football-specific cognition and that football-specific decision making better differentiates playing levels than generic cognition. These results contribute to the literature suggesting that sport-specific measures are much better predictors of skill than generic cognitive diagnostics for the purposes of talent identification.
© 2025 The Author(s). Published by Elsevier Ltd on behalf of Sports Medicine Australia. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Practical implications

- In our sample of U11–U15 youth footballers, reactive stress tolerance measured via the determination test is associated neither with football-specific decision making nor with playing level (i.e., elite vs. sub-elite).
- A generic 'FI score' assessing executive functions and anticipation could only predict a small amount of variance in sport-specific decision-making performance (4 %) beyond that possible by age (26 %).
- Football-specific decision-making performance has a higher diagnostic validity than cognitive tests using generic stimuli. The addition of these generic tests slightly improves diagnostic validity. This small added benefit should be weighed against the cost and time required.
- Practitioners should prioritise sport-specific assessments over generic cognitive tests as these are more directly relevant to performance.

1. Introduction

Research in talent identification and development often seeks to identify relevant predictors of future adult sporting performance in youth athletes. Cognitive skills, i.e., the ability to intake, process, and store information,[1] are often considered to be an important set of

* Corresponding author.
E-mail address: martin-leo.reinhard@uni-tuebingen.de (M.L. Reinhard).
Social media: @ml_reinhard (M.L. Reinhard).

https://doi.org/10.1016/j.jsams.2025.01.010
1440-2440/© 2025 The Author(s). Published by Elsevier Ltd on behalf of Sports Medicine Australia. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

predictors of talent[2] and performance.[3] Proponents of the *cognitive component skills* approach[4] argue that skilled athletes have better overall generic cognitive skills (e.g., executive functions) than lesser-skilled or novice athletes, and therefore use generic tests of cognitive skills with common stimuli (e.g., arrows, coloured circles) to test expertise differences. Advocates of the *expert performance* approach[5] contend that skilled athletes outperform others only or much better in tests of sport-specific cognitive skill (e.g., decision making). Consequently, they advocate the utilisation of sport-specific stimuli for testing cognitive skills (e.g., video clips from the respective sport). Whilst some meta-analyses and systematic reviews support the usefulness of generic approaches,[6,7] others support domain-specific approaches[8] to differentiate skilled athletes from others. The terms *generic* and *sport-specific* can refer to the extent to which both the stimuli and the participants' response correspond to those in the natural environment.[9] In this study, we focus on the stimuli as one means of increasing the representativeness of cognitive tasks[10] by using stimulus–response tests in which the presented stimuli represent either generic or football-specific stimuli to test cognitive skills.

Across sports, an argument has been put forward that generic cognitive skills such as the core executive functions may serve as the foundation for higher cognitive functioning skills such as problem-solving[11] and sport-specific cognitive skills such as decision making.[12] This may also be the case in football given that players need to perceive the environment, anticipate the movements of the ball and players, and decide on the best possible course of action. Therefore, it seems plausible to assume that generic cognitive skills may be associated with football-specific decision-making skills. Some recent evidence suggests that both generic and football-specific cognitive skills may be related to skill level in the sport.[13,14]

Decision making is a sport-specific cognitive skill that refers to the ability to find a suitable solution for a given situation, often with tight time-constraints and performed under high levels of pressure and complexity.[3] In football, decision-making skills are considered crucial predictors of talent[2] that develop with age.[15,16] In a similar way, sport-specific *anticipation*, defined as the ability to foresee what is the most likely outcome of an event before it occurs,[17] might be a relevant predictor of talent in football.[2] Most studies use a temporal and/or spatial occlusion paradigm with sport-specific video sequences to test anticipation.[18]

Generic tests of core *executive functions* typically include tests of working memory capacity, inhibition, and cognitive flexibility. These three concepts refer to self-regulatory processes and are related, yet are clearly distinct from each other and develop during childhood and adolescence.[19] In football, the expert-novice paradigm has often been applied to study differences in executive functions across players of different skill levels. Regarding working memory, expertise differences have been found in some studies,[20] but not in others.[21] Regarding inhibition, expertise differences have been recorded in football.[21] In terms of cognitive flexibility, expertise differences in football have been found, but also that these diminish when controlling for age[16] (see Supplementary Information A for more details).

Although the usefulness of diagnostic tests of cognitive skill remains a topic of debate in sports – particularly regarding the limited ability of generic cognitive tests to predict playing level or performance[8,22] – football clubs and federations are increasingly incorporating both generic *and* football-specific tests in their methods to identify and develop talents. Two of the most popular generic tests are the *determination test*[23] included as part of the Vienna Test System, and the *NeurOlympics*[24] battery of tests. However, the extent to which generic cognitive diagnostics are associated with football-specific ability remains an ongoing debate. One recent study provides some support. Heisler et al.[14] performed a cross-sectional study of young football players that found a link between working memory capacity and the quality of the first option taken in a football-specific test of decision making (both with and without time pressure). However, the diagnostic validity of most tests commonly and increasingly used in the field remains unknown (e.g., no published reports on NeurOlympics).

Whilst football-specific cognitive skills develop not only with age but also profit from experience in the sport,[25] generic cognitive skills may develop throughout the typical process of age-related development. For example, the negatively accelerated (generic) cognitive development observed in youth footballers has been shown to correspond to the typical developmental trajectory of the general youth population.[26] Also, it has been found that the relationship between executive functions and football performance is impacted by age,[27] and even diminishes when controlling for age.[16] Therefore, including chronological age as a confounding variable is crucial in studies of cognitive test performance in children.[8]

The aim of the present study was to establish the degree to which commonly used diagnostic tests of generic cognitive skills are associated with football-specific skills in youth football. To achieve that aim, we first sought to establish the association between performance on common tests of generic cognition (the determination test and NeurOlympics) and performance on a football-specific test of decision making. Second, we sought to establish the diagnostic validity of each of the tests by comparing the performance on those tests of players competing at the elite and sub-elite levels. For the first objective, we hypothesised that performance on both tests of generic cognitive skills would be positively associated with performance on the football-specific test of decision making. For the second objective, we hypothesised that the football-specific test of decision making would better differentiate players of higher skills in football (i.e., show higher diagnostic validity) than the tests of generic cognitive skills. Due to the ongoing debate regarding the relevance of generic and football-specific cognitive skills for football performance, we also investigated whether adding generic cognitive skills to football-specific cognitive skills would increase the diagnostic validity. Moreover, our study design allowed us to perform a comparative evaluation of the relative contributions of generic and sport-specific cognitive test to playing level in the same cohort of players.

2. Methods

A total of $N = 120$ male youth football players competing in the U11 to U15 age groups participated in the study. Players stemmed from an elite football academy ($n = 60$) and from a youth department of a semi-professional (4th league) football club ($n = 60$). Players in these age groups play in different playing positions throughout the season, therefore the playing position was not considered the inclusion criteria (only outfield players participated). No female players were recruited due to an absence of under-age female teams at the respective clubs and the inherent differences in the playing levels of boys and girls of the same age. Data from players who did not complete all tests were removed from all analyses ($n = 10$). The research project was conducted in accordance with the Declaration of Helsinki and approved by the institutional ethics committee of the first and last author.

We used existing cognitive diagnostics to assess generic and football-specific cognition.

- *Determination test*. The determination test of the Vienna Test System (SCHUHFRIED GmbH, Austria) was used to measure reactive stress tolerance and the related reaction speed.[23] Participants completed the 'S1: Adaptive Short' version of the test that lasts 4 min. Participants needed to respond as quickly as possible to three types of stimuli: (a) a circle filled with one of five colours, presented amongst ten circles arranged in two rows of five, by pressing respective buttons on the keyboard; (b) two acoustic signals (100 Hz low tone and 200 Hz high tone) by pressing a corresponding black or grey rectangular button on the keyboard, and (c) two visual grey, rectangular signals (bottom left, bottom right) by pressing respective foot pedals (see Supplementary Information B for more information). Stimuli were presented on the screen in a standardised, fixed order, with the speed of

presentation increasing or decreasing as a function of the participants' performance, thus putting the participant under permanent (time) pressure. The number of correct responses and the median reaction time in milliseconds were recorded as the outcome variables.[26]

- *NeurOlympics*. The NeurOlympics test battery (BrainsFirst BV, Netherlands) is designed to measure the cognitive functioning of individuals using four generic tests that assess working memory, anticipation, cognitive control, and attention (see Supplementary Information B for more information). BrainsFirst calculates a "Football Intelligence score" (FI score) that is said to predict an individual's football ability based on their performance on the generic tests of cognitive function. This FI score takes the weighted performance on an individual across all tests and calculates a percentile rank that compares the performance of that individual to that of a database of football players of the same age. The exact weights and calculation of the FI score as well as the size of the respective normative group have not been made public. The test battery lasts about 45 min in total, including a short self-paced break between each test. The FI score served as the outcome variable.
- *Football-specific test of decision making*. A 360° video decision-making test was used to measure the football-specific decision-making ability of the participants. The test involves 54 video clips of 6 vs. 6 game sequences on a half field of a U19 youth academy team filmed from the position of a central midfield player, each clip lasting approximately 10 s.[28] The 360° videos were presented in a Head Mounted Display (Vive Pro, HTC Corporation, Taiwan) allowing the players to look around freely. Participants were asked to imagine that they were the central midfield player who received the ball at the end of each video clip and to decide what they would do in that situation. The ball when passed to them was marked with one of three possible overlaying colours for 500 ms. The video turns grey when the ball arrives at the player's position. Participants were then asked to verbally answer within 2 s (a) what they would do on the pitch, choosing from five possible actions (keep the ball, pass to the left defender, pass to the central defender, pass to the right defender, or pass to the striker), and (b) what was the colour overlaying the ball. The trial was counted as correct only if both statements were correct. The correct option for every video was rated and agreed upon by three football coaches holding an UEFA Pro-Level Licence. The test consisted of two blocks. Before each block participants completed three practice videos. In the first block, the participants received the ball from the striker. In the second block, the participants received the ball from the left defender (see Supplementary Information B for more information). The test took approximately 35 min to complete. The response accuracy (in %) served as the outcome variable.

Participants performed the tests in a randomised order at the beginning of the 2022/23 football season. For the *determination test* and *360° video decision-making test*, players were tested individually by one of three trained experimenters. For the *NeurOlympics* test, teams were tested as a group.

Data were analysed using IBM SPSS version 28.0.1.1 and R version 4.2.2. As is customary in expertise research[29] due to the limited number of players playing at the elite level, convenience sampling was conducted.[30] Therefore, a *post-hoc* power analysis (G*Power version 3.1.9.7) was conducted and yielded an actual power of $1 - \beta = 0.72$ (Δ Adjusted $R^2 = 0.04$, $f^2 = 0.06$, $\alpha = 0.05$, $N = 110$) for the analyses of the association between generic and football-specific cognitive skills. In this cross-sectional study, age in days was controlled for to take into account the developmental trajectories of children.[19] Prerequisites for each analysis were tested and met, including lack of multicollinearity (Tolerance > 0.01, Variance Inflation Factor < 10). To test the association between the tests of generic cognitive skills and football-specific decision making (Objective 1), two hierarchical multiple regression analyses were calculated with decision-making performance as the dependent variable and the confounding variable age entered in step one, and either the reactive stress tolerance variables or FI score entered in step two. The change in explained variance compared to the nested model via adjusted R^2 served as the relevant variable. The differences in the models were tested via chi-square difference tests.

To test the diagnostic validity of the tests (Objective 2), the standardised residuals of each cognitive variable (controlled for age) were used to assess the unique contribution of the cognitive performance beyond age (see Supplementary Information C). Logistic regression models with playing level (elite vs. sub-elite) as the dependent

Table 1
Overview of sample, age, and test performances (*Mean* with *Standard deviation* in brackets) for each subgroup and for the total sample.

		n	Age	Generic			Football-specific
				Determination test of reactive stress tolerance		NeurOlympics FI Score (percentile rank)[a]	decision-making performance (%)[b]
				Reaction time (ms)[c]	Correct answers (*n*)[d]		
U11	Elite[e]	11	10.15 (0.33)	85.36 (5.87)	201.64 (21.56)	13.09 (10.67)	51.09 (10.54)
	Sub-elite[f]	9	10.09 (0.22)	95.56 (10.24)	184.78 (20.40)	3.89 (4.20)	31.88 (10.09)
	Total	20	10.12 (0.28)	89.95 (9.45)	194.05 (22.22)	8.95 (9.46)	42.45 (14.05)
U12	Elite	11	11.29 (0.19)	78.18 (5.29)	219.36 (25.50)	26.82 (25.97)	59.57 (9.74)
	Sub-elite	12	11.14 (0.21)	90.58 (7.15)	180.58 (22.38)	7.17 (7.38)	41.82 (8.83)
	Total	23	11.21 (0.21)	84.65 (8.86)	199.13 (30.63)	16.57 (20.85)	50.30 (12.82)
U13	Elite	12	12.23 (0.21)	84.58 (5.76)	202.50 (28.35)	25.25 (21.19)	59.04 (6.80)
	Sub-elite	8	12.22 (0.32)	86.25 (11.50)	189.63 (20.40)	15.50 (12.41)	47.34 (8.23)
	Total	20	12.23 (0.25)	85.25 (8.28)	197.35 (25.70)	21.35 (18.46)	54.36 (9.29)
U14	Elite	12	13.14 (0.24)	81.08 (4.66)	211.00 (22.50)	19.50 (17.72)	57.98 (7.37)
	Sub-elite	12	13.46 (0.32)	75.25 (6.34)	227.25 (24.53)	17.25 (18.62)	57.37 (14.80)
	Total	24	13.30 (0.32)	78.17 (6.20)	219.13 (22.87)	18.38 (17.81)	57.68 (11.44)
U15	Elite	11	14.12 (0.56)	76.09 (6.22)	225.55 (22.14)	18.82 (16.13)	67.45 (7.41)
	Sub-elite	12	14.32 (0.34)	72.17 (7.48)	244.92 (32.39)	10.42 (11.60)	58.62 (10.40)
	Total	23	14.22 (0.46)	74.04 (7.04)	235.65 (29.08)	14.43 (14.28)	62.84 (9.97)
All players	Elite	57	12.20 (1.42)	81.12 (6.45)	211.82 (25.13)	20.75 (19.01)	59.01 (9.64)
	Sub-elite	53	12.37 (1.57)	83.13 (12.17)	207.79 (35.97)	10.89 (12.69)	48.29 (14.51)
Overall		110	12.28 (1.49)	82.09 (9.65)	209.88 (30.75)	16.00 (16.94)	53.84 (13.31)

[a] Assessed via NeurOlympics; presented in percentile rank based on age-matched football players completing the test, provided by BrainsFirst.
[b] Assessed via 360° videos, relative accuracy (% correct answers).
[c] Assessed via determination test; in milliseconds.
[d] Assessed via determination test; number of correct responses within 4 min. Boxplots displaying descriptive results can be found in the Supplementary Information H.
[e] Elite Youth Academy.
[f] Youth department of 4th league club.

Table 2
Hierarchical regression analysis results for the association between age and generic cognition with 360° video decision-making performance whilst controlling for age.

Model	Omnibus test				Predictor	Regression coefficients			
	F (df)	*p*	Adj. R^2	Δ adj. R^{2a}		*B* [95 % CI]	β	T	*p*
Model 1 (age)	38.41 (1,108)	<0.001	0.26	–	Constant	−2.32 [−20.41; 15.78]	–	–	–
					Age	4.57 [3.11; 6.04]	0.51	6.20	<0.001
Model 2 (age + FI score)	24.41 (2,107)	<0.001	0.30	0.04**	Constant	−2.38 [−19.92; 15.16]	–	–	–
					Age	4.35 [2.92; 5.77]	0.49	6.04	<0.001
					FI score	0.18 [0.05; 0.30]	0.23	2.82	0.006
Model 3 (age + reactive stress tolerance)	13.21 (3,106)	<0.001	0.25	0.00	Constant	−11.73 [−71.24; 47.78]	–	–	–
					Age	4.13 [2.31; 5.96]	0.46	4.50	<0.001
					Correct	0.06 [−0.06; 0.17]	0.13	0.99	0.326
					RT	0.04 [−0.35; 0.42]	0.03	0.19	0.851

The predictors are sorted according to the size of the standardised regression coefficients.
[a] Δ adjusted R^2 shows the increase in explained variance compared to model 1 (using solely age as a predictor).
** Significant ($p < 0.01$) difference between models 1 and 2, assessed via chi-square difference test in R.

variable were created with either the reactive stress tolerance variables, FI score, or decision-making performance as the independent variables. Further, a logistic regression model was also run with playing level (elite vs. sub-elite) as the dependent variable and the standardised residuals of all cognitive variables as the independent variables (method: stepwise forward). Differences between the models were assessed via deviance tests based on χ^2 statistics. The models were compared using Bayesian Information Criterion (BIC) and Akaike Information Criterion corrected for small sample sizes (AICc) with significance assessed through log-likelihood ratio tests. Standardised odds ratio coefficients (e^b) were computed to represent the relative change of the likelihood for being an academy player for each one standard deviation increase within the respective predictor to foster comparisons for effect sizes of individual predictors.[31] Also, univariate analyses with receiver operator characteristic (ROC) curves were calculated to examine the sensitivity and specificity for each variable.

3. Results

An overview of descriptive statistics can be found in Table 1. Bivariate correlations showed significant associations between performance on the tests of generic cognition and on the test of football-specific decision making (reactive stress tolerance, RT: $r = -0.35$, correct: $r = 0.34$; FI score: $r = 0.28$). However, there were also significant associations between the age of the participants and their performance on the tests of reactive stress tolerance (RT: $r = -0.59$, correct: $r = 0.50$) and football-specific decision making ($r = 0.51$), highlighting the need to control for age during all analyses (see Supplementary Information D and E).

To address Objective 1, hierarchical regression analyses (Table 2) showed that the FI score was associated with football-specific decision making ($p = 0.006$), but that reactive stress tolerance was not ($p \geq 0.326$). Age alone explained 26 % of the variance in the decision-making score. Further including the FI score led to a significant increase ($p = 0.006$) of the explained variance up to 30 %. The *B* score indicated that an increase of one year of age would result in an average increase in decision-making performance of more than 4 percentage points, whilst a ten percentile ranks better performance in the generic FI score, would indicate an average gain of 1.8 percentage points in the decision-making performance.

To address Objective 2, logistic regression analyses (Table 3) showed that both the football-specific decision making (Model A) and FI score (Model B) significantly differentiated the elite from sub-elite players, but that of the test of reactive stress tolerance (Model C) did not. Amongst them, Model A was clearly the best fitting one. Statistically, the standardised odds ratio showed that an increase of one standard deviation in the decision-making performance would increase the likelihood for being in the elite group almost fourfold (see the results for e^b in Table 3), whilst an increase of one standard deviation in the FI score would more than double the likelihood of being in the elite group.

Table 3
Logistic regression analysis results for the influence of *standardised residuals of cognitive test performance* on playing level (elite vs. sub-elite).

Test	Omnibus test			Correctly assigned players	Predictor	Logistic regression coefficients			
	χ^2 (df)	*p*	Nagelkerke's R^2			B	Wald	*p*	e^b (95 % CI)[a]
Model A:	32.11 (1)	<0.001	0.34	73.6 %	Constant	0.06	–	–	–
Decision making					DM	1.37	21.26	<0.001	3.93 [2.20; 7.04]
Model B:	11.01 (1)	<0.001	0.13	60.9 %	Constant	0.12	–	–	–
FI score					FI score	0.74	8.78	0.003	2.09 [1.28; 3.41]
Model C:	3.26 (2)	0.196	0.04	55.5 %	Constant	0.07	–	–	–
Reactive stress tolerance					RT	−0.38	1.89	0.169	0.68 [0.40; 1.12]
					Correct	−0.04	0.02	0.892	0.96 [0.57; 1.64]
Model ABC:	36.62 (2)	<0.001	0.38	70.0 %	Constant	0.08	–	–	–
All[b]			Δ R^2		DM	1.28	18.12	<0.001	3.61 [2.00; 6.52]
			0.04[c,*]		FI score	0.52	3.90	0.048	1.68 [1.00; 2.83]
			0.25[d,***]						
			0.34[e,***]						

[a] Standardised odds ratio.
[b] Including all four cognitive variables as independent variables (model: forward Wald). Only significant predictors in the model are listed. The predictors for each analysis are listed according to the Wald-coefficient.
[c] Indicates the change to model A.
[d] Indicates the change to model B.
[e] Indicates the change to model C.
* $p < 0.05$
*** $p < 0.001$.

Additionally, Receiver Operator Characteristic (ROC) analyses revealed that the highest probability of correct assignment to the respective playing level was found based on the residuals of the test results for decision-making performance (AUC = 0.79), followed by the FI score (AUC = 0.68) (Supplementary Information F and G).

Whilst Model ABC (combining the decision-making and FI scores) was the best fitting model in terms of AICc, AIC and log-likelihood tests, Model A (decision making only) was the most accurate in terms of percentage of correct assignment. Adding generic cognitive variables to football-specific decision making revealed decision-making performance and FI score as significant predictors, with a significant increase in explained variance of 4 percentage points compared to Model A ($p = 0.034$), and with lower AIC (121.73) and AICc (121.95) compared to Model A (AIC: 124.24, AICc: 124.35). However, the results revealed that Model ABC assigns approximately 3 percentage points less players correctly compared to Model A, and BIC analyses uncovered the lowest value for Model A with Δ BIC of 0.18 to Model ABC (see Supplementary Information F).

4. Discussion

The aim of this study was to establish the degree to which tests of generic cognitive skill are associated with sport-specific skill in youth football. There has been much discussion about the potential association between generic cognitive skills and sport performance[8,22] – and a wave of new commercially available tools to test those skills. Our findings support previous studies suggesting that cognitive skills improve with age,[15,25] and only limited evidence that they are in any way associated with skill in the sport when controlling for age.[16] Specifically, we found that youth footballers with better test results in a generic cognitive test measuring executive functions and anticipation (FI score) tended to also register better results in a football-specific test of decision making, but that the association was best explained by age. Only a small portion of the variance could be explained (i.e., 4 %) after age was controlled for. Additionally, the playing level of players was best predicted by performance on the sport-specific test of decision-making, with performance on tests of executive function and anticipation (FI score) making only a small improvement in variance explained, and the test of reactive stress tolerance providing no improvement. Overall, these results support the idea that sport-specific tests of perceptual-cognitive skill are better discriminators of playing level in youth footballers, and that generic tests of cognitive skill make only a limited contribution beyond the insights provided by sport-specific assessments.

The results demonstrate that both the generic FI score (for executive functioning and anticipation) *and* football-specific decision-making performance together can significantly differentiate between two playing levels in youth football players. Yet, the football-specific decision-making test assigned more players correctly and explains significantly more variance (34 %) than the FI score (13 %), therefore, having the highest diagnostic validity. This result is in alignment with what would be expected,[8] and in particular is likely due to the lesser degree to which the stimuli in the NeurOlympic test represents that experienced in the competition environment.[9,10] This conclusion is further supported by the fact that the test of generic reactive stress tolerance failed to differentiate between playing level, confirming doubts about the usefulness of this generic cognitive test for talent identification purposes.[8,26] The fact that the FI score demonstrated (some) diagnostic validity, whereas the reactive stress tolerance scores did not, may suggest that executive functions and generic anticipation do make some limited form of contribution to football-specific cognitive skills in youth football whereas reactive stress tolerance does not. Alternatively, the result could be explained by the relative differences in the length of the test (reactive stress tolerance: 4 min, FI score: ca. 45 min).

There was some suggestion that a model that combined the FI score with football-specific decision-making was better able to predict playing level than when using a model with the test of decision making alone. The combined model increased the explained variance in skill level from 34 % to 38 % (Table 3). Depending on the statistical analyses, either the combined model (i.e., based on AIC and AICc) or the model with decision making alone (based on % of correctly assigned players) was best. From an applied perspective, this study builds on previous recommendations[8] to prioritise sport-specific (decision making) assessments over generic cognitive tests. Additionally, a consideration of the cost–benefit trade-off may be necessary to decide whether an additional explanation of 4 percentage points in variance warrants the additional time and resources necessary to test generic cognitive skills in addition to football-specific cognitive tests (such as the decision-making performance).

Given that the FI score used in the test of NeurOlympics is a composite score across four different constructs, it is not possible to conclude what might be individual contributions of (small) extent to which each of the generic cognitive skills (working memory, anticipation, cognitive control, attention) might be linked to football-specific skill. It has previously been found in cross-sectional studies that, amongst executive functions, especially working memory is linked to football-specific decision making in young football players,[15] and that working memory and cognitive flexibility (but not selective attention and inhibition) are associated with coach-rated game intelligence in a sample of youth and adult football players.[32] However, neither this nor other cross-sectional studies allow for an empirical conclusion regarding the cause–effect relationship.

It is important to be clear about the operationalisation of the constructs being measured in this and in other studies of cognitive skills. Concerns exist about the *task impurity* of tests of executive functions in particular,[22] but this can also be extended to other tests. The *Jingle-Jangle-Fallacy* represents the (sometimes mistaken) belief that "scales with the same label reflect the same construct […] [and that] scales with different labels measure different constructs".[33] In this study, the "Football Intelligence" score was based on performance on tests using generic stimuli to assess working memory, anticipation, cognitive control, and attention. The term "Football Intelligence", however, may imply that football-specific stimuli (and answers) have been used or required. Also, the *determination test* has been used to assess reactive stress tolerance in this study, but in other studies also as a measure for cognitive flexibility[16] and other constructs.[34] Finally, decision-making (competence) has been conceptualised in various ways differing also along the answer requisite.[9] Therefore, it is vital to be clear and transparent about the operationalisation of measured constructs and in communicating the uncertainty and alterability of results to relevant stakeholders.[35] This holds true for all cognitive tests used in this study, where we have drawn on the test manual, and elsewhere.

5. Limitations

This study is amongst the first to investigate the association between generic and football-specific cognitive skills. We consider as strengths of the study the elite sample and use of three tests that are increasingly commonly used in the applied field. The results may serve as a building block for further empirical studies, as limitations remain. First, convenience sampling was necessary due to the limited access to elite youth footballers. Future studies may use larger and more heterogeneous (e.g., regarding sex) samples to increase statistical power and to gain more insight into sex-related differences. This may be possible if clubs and federations that use cognitive diagnostics as regular part of their player testing and monitoring are willing to share their results with the scientific community. Second, we were not able to control for other potentially confounding variables such as emotional or physical characteristics.[22] We addressed this by using three trained experimenters to conduct the testing using standardised instructions and by testing the players in the same time range (between 5 pm and 8 pm) in a rested state. Additionally, we evaluated whether the weekly hours spent gaming are associated with the cognitive results (which was not the case, see Supplementary Information I). Results may have been

affected by motivational factors, as participants received real-time feedback on their performance in *NeurOlympics* (but not the other tests). This is unusual in typical cognitive performance tests but may better represent on-field performance where immediate feedback about success is readily available. Also, we did not control for positional effects as players in younger age groups play across several positions throughout a season. Future studies with older ages groups (e.g., U17, U19 and adult samples) should include information about position, especially (but not solely) for sport-specific cognitive tests. Third, the 360° video decision-making test (which has been evaluated for older age groups[28]) made use of verbal responses within a specific time-period (2 s) instead of motor responses, hence, limiting the representativeness of the design. The authors of a recent framework for decision-making diagnostics and training argue for the consideration of the perception-action cycle and the importance of representativeness.[36] In our study, we used a decision-making test with football-specific stimuli (i.e., 360° videos) which is one possibility to increase representativeness in cognitive tests.[10] In fact, the research design employed in this study required a generic response for all tests to elicit performance differences in cognitive functions regardless of motor skills. For example, the 360° video decision-making test solely tests the perceptual-cognitive decision-making skill without consideration of the technical skill of this player.[37] Incorporating context-specific action responses (e.g., passing[38]) instead of verbal responses may be one further strategy to enhance the representativeness of decision-making tests. Thus, future studies may include (standardised) motor responses[9] and define a "window of opportunity"[36] in which a given decision may be correct. Alternatively, on-field decision-making quality may be assessed, e.g., via small-sided games.[39]

Finally, it is important to note that using the current playing level of players as a criterion only provides limited information on inter-individual differences and neglects prognostic validity.[40] The latter has been evaluated for the 360° video decision-making test in a sample of U17 and U19 football players,[28] but should also be assessed in younger age groups and across a wider range of cognitive skills in future research.

6. Conclusion

In conclusion, the current study found a partial association between performance on tests of generic and football-specific cognitive skills in youth football players. However, the football-specific test of cognitive skill differentiated skill level much better than the different generic tests. This adds to the literature warranting caution in the use of generic cognitive diagnostics for talent identification and selection. We found an age-independent association between some generic and football-specific measures of cognition, with a generic 'Football intelligence score' designed to evaluate executive functions and generic anticipation being associated with football-specific decision-making performance. Regarding diagnostic validity, football-specific decision making was found to have a higher explanatory power than the tests of generic cognition when differentiating youth elite from sub-elite players. Yet, the addition of the FI score did marginally improve the diagnostic validity. However, caution is required when seeking to generalise these results, because a cross-sectional instead of longitudinal study design has been applied. In the quest for more definite answers, future studies should continue to consider age as a covariate when assessing the diagnostic validity of cognitive tests in sport settings, employ designs that allow conclusions about cause–effect relationships, and further focus on applied cognitive diagnostics that are used by clubs and players as tools for decisions about selection and/or player development.

CRediT authorship contribution statement

Martin L. Reinhard: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Visualization, Writing – original draft. **David L. Mann:** Writing – review & editing. **Oliver Höner:** Conceptualization, Methodology, Funding acquisition, Resources, Supervision, Project administration, Writing – review & editing.

Confirmation of ethical compliance

The players took part in this study voluntarily. This study was conducted in accordance with the ethical standards of the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The studies involving human participants were reviewed and approved by the Ethics Committee at the Faculty of Economics and Social Sciences at the University of Tübingen (A2.5.4-181_ns) and the youth academy and the local club approved the usage of the data. Written informed consent to participate in this study was obtained from the players' legal guardian/next of kin from all participants.

Funding information

The DFB-Akademie supported this research by covering the costs for the licences for NeurOlympics and by supplying hardware and software for the determination test. The company offering NeurOlympics (BrainsFirst) provided a discount for scientific endeavours. All external partners had no influence on the study design, execution, reporting and publishing of results.

Declaration of interest statement

The authors declare that they have no competing interests. Martin Leo Reinhard works as a sport psychologist for the youth academy of which players have been recruited. This employment had no influence on the analysis and reporting of results of the study.

Acknowledgements

The authors thank all participating players and both their parents and coaches for the time, effort, and support. Also, the authors thank Viktoria Vochatzer and Nils Jauch for their support during data collection. The authors thank Chris Herr and Daniel Brinkmann from the DFB-Akademie for supplying hardware and licences for testing. The authors are grateful for the support and feedback from the colleagues from the department for sport psychology and research methods during manuscript execution.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jsams.2025.01.010.

References

1. Lachman R, Lachman JL, Butterfield EC. *Cognitive psychology and information processing: an introduction*, Psychology Press, 2015.
2. Williams AM, Ford PR, Drust B. Talent identification and development in soccer since the millennium. *J Sports Sci* 2020;38(11–12):1199-1210. doi:10.1080/02640414.2020.1766647.
3. Höner O, Larkin P, Leber T et al. Talent identification and development in sport. *Sport Exerc Psychol Theory Appl* 2023:549-581. [Springer].
4. Nougier V, Stein J-F, Bonnel A-M. Information processing in sport and "orienting of attention". *Int J Sport Psychol* 1991;22:307-327.
5. Ericsson KA, Smith J. *Toward a general theory of expertise: prospects and limits*, Cambridge University Press, 1991.
6. Logan NE, Henry DA, Hillman CH et al. Trained athletes and cognitive function: a systematic review and meta-analysis. *Int J Sport Exerc Psychol* 2023;21(4):725-749. doi:10.1080/1612197X.2022.2084764.
7. Scharfen H, Memmert D. Measurement of cognitive functions in experts and elite athletes: a meta-analytic review. *Appl Cogn Psychol* 2019;33(5):843-860. doi:10.1002/acp.3526.
8. Kalén A, Bisagno E, Musculus L et al. The role of domain-specific and domain-general cognitive functions and skills in sports performance: a meta-analysis. *Psychol Bull* 2021;147(12):1290-1308. doi:10.1037/bul0000355.
9. Travassos B, Araújo D, Davids K et al. Expertise effects on decision-making in sport are constrained by requisite response behaviours – a meta-analysis. *Psychol Sport Exerc* 2013;14(2):211-219. doi:10.1016/j.psychsport.2012.11.002.

10. Hadlow SM, Panchuk D, Mann DL et al. Modified perceptual training in sport: a new classification framework. *J Sci Med Sport* 2018;21(9):950-958. doi:10.1016/j.jsams.2018.01.011.
11. Diamond A. Executive functions. *Annu Rev Psychol* 2013;64(1):135-168. doi:10.1146/annurev-psych-113011-143750.
12. Memmert D, Roca A. Tactical creativity and decision making in sport, In: Mark WA, Jackson Robin C, eds. *Anticip. Decis. Mak. Sport*, 1st ed., New York, NY, Routledge, 2019. p. 201-214.
13. Murr D, Feichtinger P, Larkin P et al. Psychological talent predictors in youth soccer: a systematic review of the prognostic relevance of psychomotor, perceptual-cognitive and personality-related factors. *PLoS One* 2018;13(10):e0205337. doi:10.1371/journal.pone.0205337.
14. Ivarsson A, Kilhage-Persson A, Martindale R et al. Psychological factors and future performance of football players: a systematic review with meta-analysis. *J Sci Med Sport* 2020;23(4):415-420. doi:10.1016/j.jsams.2019.10.021.
15. Heisler SM, Lobinger BH, Musculus L. A developmental perspective on decision making in young soccer players: the role of executive functions. *Psychol Sport Exerc* 2023;65:102362. doi:10.1016/j.psychsport.2022.102362.
16. Radke L, Mertens A, Spielmann J et al. Being ahead of the game—the association between executive functions and football performance in high-level football players. *Ger J Exerc Sport Res* 2023:1-13.
17. Gonçalves E, Gonzaga ADS, Cardoso FDSL et al. Anticipation in soccer: a systematic review. *Hum Mov* 2015;16(2). doi:10.1515/humo-2015-0032.
18. Loffing F, Cañal-Bruland R. Anticipation in sport. *Curr Opin Psychol* 2017;16:6-11. doi:10.1016/j.copsyc.2017.03.008.
19. Karr JE, Areshenkoff CN, Rast P et al. The unity and diversity of executive functions: a systematic review and re-analysis of latent variable studies. *Psychol Bull* 2018;144(11):1147-1185. doi:10.1037/bul0000160.
20. Vestberg T, Reinebo G, Maurex L et al. Core executive functions are associated with success in young elite soccer players. *PLoS One* 2017;12(2):e0170845. doi:10.1371/journal.pone.0170845.
21. Verburgh L, Scherder EJA, Van Lange PAM et al. Executive functioning in highly talented soccer players. *PLoS One* 2014;9(3):e91254. doi:10.1371/journal.pone.0091254.
22. Furley P, Schütz L-M, Wood G. A critical review of research on executive functions in sport and exercise. *Int Rev Sport Exerc Psychol* 2023;0(0):1-29. doi:10.1080/1750984X.2023.2217437.
23. Neuwirth W, Benesch M. *Manual DT: determination test, version 33.00,* SCHUHFRIED GmbH Mödling Austria, 2007.
24. Alkmaar AZ. AZ and BrainsFirst extend agreement. AZ and BrainsFirst Extend Agreement. Available at: https://www.az.nl/en/nieuws/az-and-brainsfirst-extend-agreement. Accessed 29 November 2023.
25. Klatt S, Smeeton NJ. Processing visual information in elite junior soccer players: effects of chronological age and training experience on visual perception, attention, and decision making. *Eur J Sport Sci* 2022;22(4):600-609. doi:10.1080/17461391.2021.1887366.
26. Beavan A, Spielmann J, Mayer J et al. The rise and fall of executive functions in high-level football players. *Psychol Sport Exerc* 2020;49:101677. doi:10.1016/j.psychsport.2020.101677.
27. Heilmann F, Wollny R, Lautenbach F. Inhibition and calendar age explain variance in game performance of youth soccer athletes. *Int J Environ Res Public Health* 2022;19(3):1138. doi:10.3390/ijerph19031138.
28. Höner O, Dugandzic D, Hauser T et al. Do you have a good all-around view? Evaluation of a decision-making skills diagnostic tool using 360° videos and head-mounted displays in elite youth soccer. *Front Sports Act Living* 2023;5:190.
29. Schweizer G, Furley P. Reproducible research in sport and exercise psychology: the role of sample sizes. *Psychol Sport Exerc* 2016;23:114-122.
30. Lakens D. Sample size justification. *Collabra Psychol* 2022;8(1):33267.
31. Höner O, Murr D, Larkin P et al. Nationwide subjective and objective assessments of potential talent predictors in elite youth soccer: an investigation of prognostic validity in a prospective study. *Front Sports Act Living* 2021;3:638227. doi:10.3389/fspor.2021.638227.
32. Scharfen H, Memmert D. Fundamental relationships of executive functions and physiological abilities with game intelligence, game time and injuries in elite soccer players. *Appl Cogn Psychol* 2021;35(6):1535-1546.
33. Marsh HW. Sport motivation orientations: beware of jingle-jangle fallacies. *J Sport Exerc Psychol* 1994;16(4):365-380.
34. Ong NCH. The use of the Vienna Test System in sport psychology research: a review. *Int Rev Sport Exerc Psychol* 2015;8(1):204-223. doi:10.1080/1750984X.2015.1061581.
35. Kilger M, Blomberg H. Governing talent selection through the brain: constructing cognitive executive function as a way of predicting sporting success. *Sport Ethics Philos* 2020;14(2):206-225.
36. Janssen T, Müller D, Mann DL. From natural towards representative decision making in sports: a framework for decision making in virtual and augmented environments. *Sports Med* 2023;53(10):1851-1864. doi:10.1007/s40279-023-01884-3.
37. Bruce L, Farrow D, Raynor A et al. But I can't pass that far! The influence of motor skill on decision making. *Psychol Sport Exerc* 2012;13(2):152-161. doi:10.1016/j.psychsport.2011.10.005.
38. Murr D, Larkin P, Höner O. Decision-making skills of high-performance youth soccer players: validating a video-based diagnostic instrument with a soccer-specific motor response. *Ger J Exerc Sport Res* 2021;51(1):102-111. doi:10.1007/s12662-020-00687-2.
39. Ehmann P, Beavan A, Spielmann J et al. Perceptual-cognitive performance of youth soccer players in a 360°-environment–an investigation of the relationship with soccer-specific performance and the effects of systematic training. *Psychol Sport Exerc* 2022;61:102220.
40. Bergkamp TLG, Niessen ASM, Den Hartigh Ruud JR et al. Methodological issues in soccer talent identification research. *Sports Med* 2019;49(9):1317-1335. doi:10.1007/s40279-019-01113-w.